

QRNG Randomness Audit — NIST SP 800-22, all 15 tests

Source: data/random_100_mb.bin | 100 sequences per test (70 for Random Excursions) | 188 test rows (148 are NonOverlappingTemplate sub-tests)

15/15 tests pass

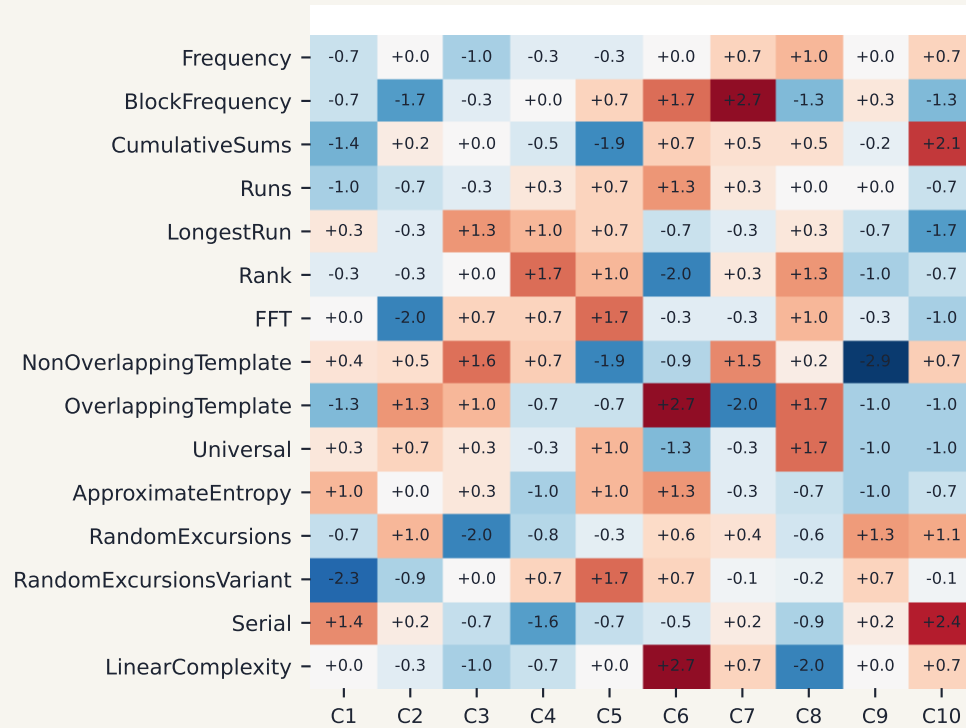
0.0192 lowest uniformity p
(fail < 0.0001)

96% lowest pass rate
(min ≈ 96%)

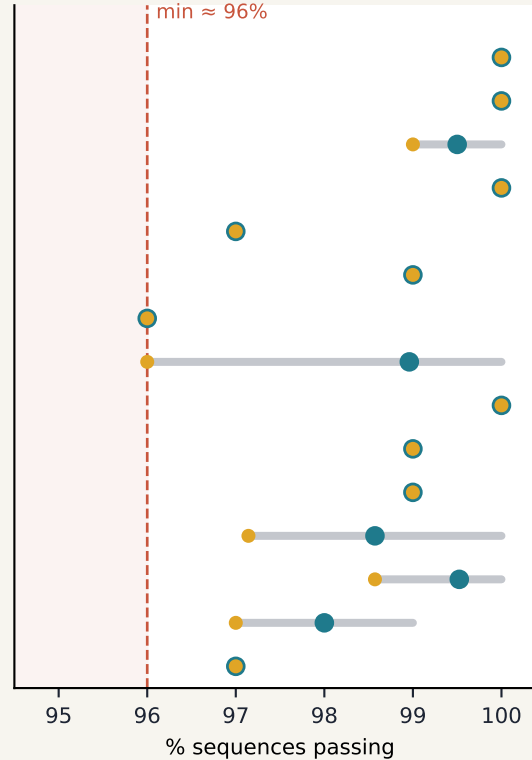
99.0% mean pass rate

0 p-values < 0.01
(of 188)

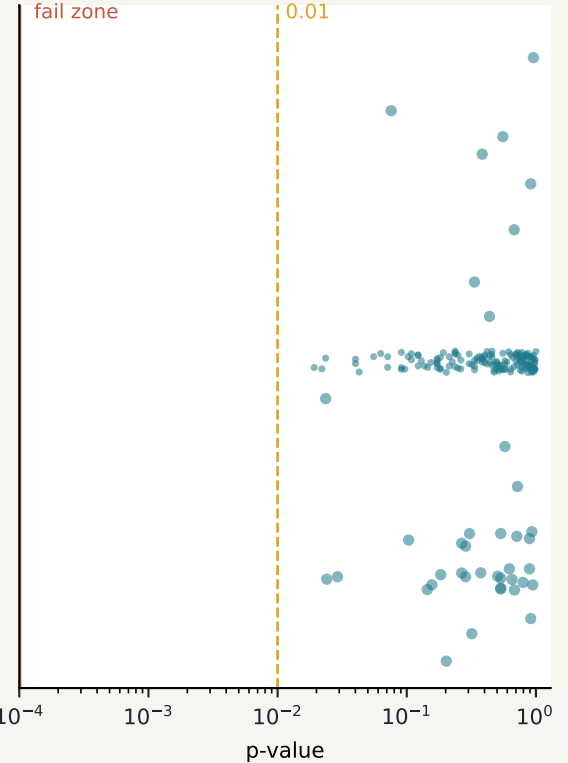
A Bin deviation (z-score, 0 = ideal)



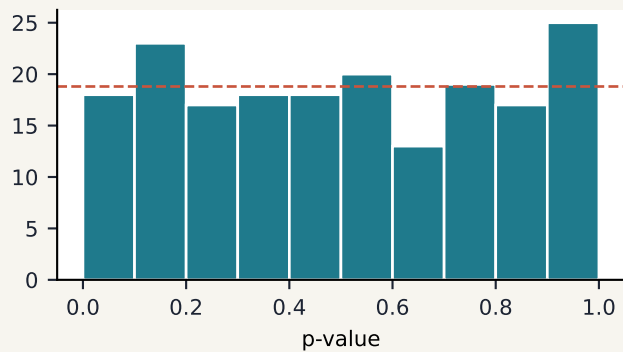
B Pass rate (teal=mean, gold=worst)



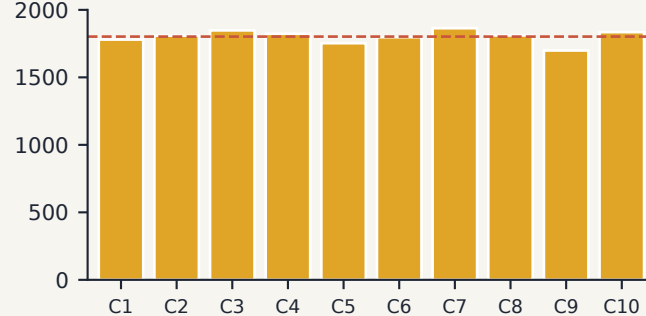
C Uniformity p-values (log scale)



D All 188 p-values: uniform spread



E Pooled C1-C10 counts (all 188 rows)



Verdict: PASS — all 15 NIST tests accept the data as random.

- Lowest pass rate 96% sits exactly on the 96% line; expected occasionally across 188 rows.
- Pooled bins chi-square = 11.7 (9 dof; crit. 16.9).
- No p-value comes near the 0.0001 failure line.

Note: NonOverlappingTemplate = 148 sub-tests;
Random Excursions (Variant) = 8 (18).